

MSTAR PRACTICE WORKSHEET**Lesson 3-5: Compare Ratio Relationships**

Grade 6 Mathematics · Standard 6.AT.3

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: B

Chef A uses 2 cups of cheese for every 8 crackers. Chef B uses 3 cups of cheese for every 9 crackers. Who uses more cheese per cracker?

- ☐ A Chef A
- ☒ B Chef B
- ☐ C They use the same amount
- ☐ D Cannot determine

Chef A: $2 \div 8 = 0.25$ cups per cracker. Chef B: $3 \div 9 \approx 0.33$ cups per cracker. Chef B uses more cheese per cracker.

Question 2 SELECTED RESPONSE — CORRECT: B

Which ratio is greater: 3:4 or 5:8?

- ☐ A 5:8
- ☒ B 3:4
- ☐ C They are equal
- ☐ D Cannot determine

Convert to the same denominator: $3:4 = 6:8$. Since $6:8 > 5:8$, the ratio 3:4 is greater.

Question 3 **SELECTED RESPONSE — CORRECT: A**

Which lemonade recipe is more lemony? Recipe A: 2 lemons for 6 cups of water. Recipe B: 3 lemons for 7 cups of water.

- ☒ A Recipe B
- ☐ B Recipe A
- ☐ C They are the same
- ☐ D Cannot determine

Recipe A: $2 \div 6 \approx 0.33$ lemons per cup. Recipe B: $3 \div 7 \approx 0.43$ lemons per cup. Since $0.43 > 0.33$, Recipe B is more lemony.

Question 4 **SELECTED RESPONSE — CORRECT: B**

Which is the better deal: 4 apples for \$3 or 6 apples for \$5?

- ☐ A 6 apples for \$5
- ☒ B 4 apples for \$3
- ☐ C They cost the same
- ☐ D Cannot determine

Deal 1: $\$3 \div 4 = \0.75 per apple. Deal 2: $\$5 \div 6 \approx \0.83 per apple. $\$0.75 < \0.83 , so 4 apples for \$3 is the better deal.

Question 5 **SELECTED RESPONSE — CORRECT: C**

Store A sells 3 mangoes for \$6. Store B sells 5 mangoes for \$9. Which store offers a better price per mango?

- ☐ A Store A
- ☐ B Same price
- ☒ C Store B
- ☐ D Cannot determine

Store A: $\$6 \div 3 = \2.00 per mango. Store B: $\$9 \div 5 = \1.80 per mango. Store B has the better (lower) price per mango.

Question 6

SELECTED RESPONSE — CORRECT: B

Chef Academy's snack bar sells 3 mini quiches for \$6 or 5 mini quiches for \$8. Which is the better deal per quiche?

- ☐ A 3 quiches for \$6
- ☒ B 5 quiches for \$8
- ☐ C They cost the same
- ☐ D Cannot determine

3 for \$6 is $\$6 / 3 = \2.00 per quiche. 5 for \$8 is $\$8 / 5 = \1.60 per quiche. $\$1.60 < \2.00 , so 5 quiches for \$8 is the better deal.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7**TWO-PART QUESTION****Part A — correct answer: D**

Car A travels 180 miles on 6 gallons. Car B travels 225 miles on 9 gallons. Which car gets better gas mileage?

- ☐ A Car B, at 25 miles per gallon
- ☐ B Car B, because it traveled farther
- ☐ C They are equal
- ☒ D Car A, at 30 miles per gallon

Car A: $180 \div 6 = 30$ miles per gallon. Car B: $225 \div 9 = 25$ miles per gallon. $30 > 25$, so Car A gets better mileage.

- **A:** 25 miles per gallon is Car B's rate and it is computed correctly, but 30 is the larger rate.
- **B:** Car B did travel farther, but it burned more gas doing it. Compare miles PER GALLON.
- **C:** 180 divided by 6 is 30 and 225 divided by 9 is 25, so the two rates differ.

Part B — correct answer: C

Why is comparing total miles alone NOT enough to decide which car is more efficient?

- ☐ A Total miles is never a useful measurement.
- ☐ B The car with fewer miles is always more efficient.
- ☒ C The cars used different amounts of gas, so the totals are not comparable until each is expressed per gallon.
- ☐ D Efficiency depends only on the size of the gas tank.

Car B traveled farther but also used more gas. Converting each to a unit rate puts both on the same per-gallon basis, which is what makes them comparable.

Question 8

WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: Comparing Totals Instead of Rates

Jaden compared two paint sizes: a 3-gallon can for \$54 and a 5-gallon can for \$85. He wrote: 'The 3-gallon can is the better deal because \$54 is less than \$85.'

Explain Jaden's misconception and determine which can is actually the better deal.

Model answer: $54 \div 3 = \$18$ per gallon; $85 \div 5 = \$17$ per gallon. The 5-gallon can is the better deal. Jaden compared totals for different amounts of paint; only the price per gallon compares them fairly.

2 points	Student computes both unit prices (\$18 per gallon and \$17 per gallon), identifies the 5-gallon can as the better deal, and explains that a lower total price does not mean a lower price per gallon.
1 point	Student identifies the 5-gallon can but does not show or explain the unit prices.
0 points	Student agrees with Jaden or gives an unrelated response.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9

WRITTEN RESPONSE · 2 POINTS

Store A sells 5 notebooks for \$12. Store B sells 8 notebooks for \$20. Without calculating, a student says Store B must be cheaper because you get more notebooks. Is the student's reasoning valid? Find the unit rates to prove your answer.

Model answer: The student's reasoning is not valid — getting more items does not mean a lower price per item. Store A: $\$12 \div 5 = \2.40 per notebook. Store B: $\$20 \div 8 = \2.50 per notebook. Store A is actually cheaper per notebook, even though Store B offers more notebooks per purchase.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.